Task 9 & 10 Report - Intelligent Fraud Detection System

1. System integration

FraudPipeline (src/pipeline.py) integrates validation, feature engineering, two models (Keras DNN, logistic regression), SHAP explanations, a nearest-neighbour trust score, human-in-the-loop routing and report generation. A Streamlit app (app.py, pages/) exposes it through six pages and only calls src/. Dataset: 5,389 transactions, 49.3% fraud; split 70/15/15 stratified (3772/808/809); preprocessor fit on train only.

Key finding. Both models score close to chance (test ROC-AUC 0.524 DNN, 0.510 LR). A 5-fold gradient-boosting check on the raw data gives ~0.51 as well. The dataset shows no learnable fraud signal (fraud rate is ~49% in every category, and the account numbers and descriptions are unique per row, which suggests simulated data). The system demonstrates integration, explainability, testing and governance; it does not demonstrate real fraud-detection skill.

2. Deliverables inventory

Deliverable	Path
Pipeline	src/pipeline.py
Features / data / training / evaluation	src/features.py, src/data.py, src/train.py, src/evaluate.py
App	app.py, pages/, src/ui.py, .streamlit/config.toml
Models	models/fraud_lr_model.pkl, models/fraud_dnn_model.keras, models/preprocessor.pkl, models/metadata.json
Metrics and CSVs	reports/metrics.json, cost_threshold.csv, latency.json, shap_stability.json, fairness_gender.csv, fairness_age.csv, robustness.csv, drift_baseline.csv, deployment_readiness.csv, testing_report.csv
Figures	reports/figures/
Tests	tests/
Documentation	docs/SRS.md, SDD.md, USER_MANUAL.md, RESPONSIBLE_AI_REPORT.md, DEPLOYMENT_GUIDE.md, TESTING_REPORT.md, FINAL_REPORT.md
Notebook	notebooks/01_eda_and_training.ipynb
Deployment	Dockerfile, scripts/run_all.sh

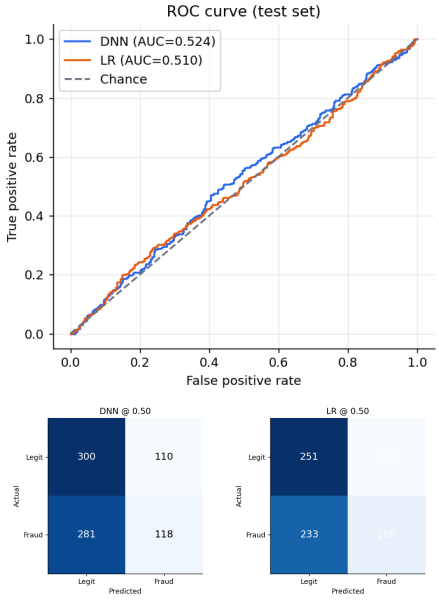
3. Functional testing

13/13 automated tests pass (valid score, negative amount rejected, missing field rejected, probabilities in [0,1], recommendation in allowed set, non-empty report, batch of 20, batch-vs-single parity, latency). Details: docs/TESTING_REPORT.md.

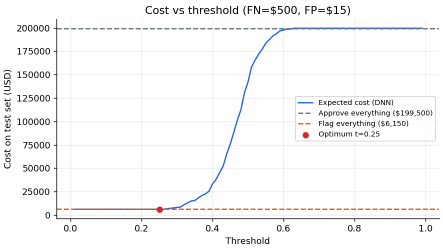
4. Performance evaluation

Metric	DNN (threshold 0.50)	Logistic regression (0.50)
ROC-AUC	0.524	0.510
PR-AUC	0.508	0.506
F1	0.376	0.459
Precision	0.518	0.511
Recall	0.296	0.416
Accuracy	0.517	0.515
Flag rate	28.2%	40.2%
Confusion matrix (TN/FP/FN/TP)	300/110/281/118	251/159/233/166

Test set: 809 transactions (49.3% fraud). Source: reports/metrics.json.



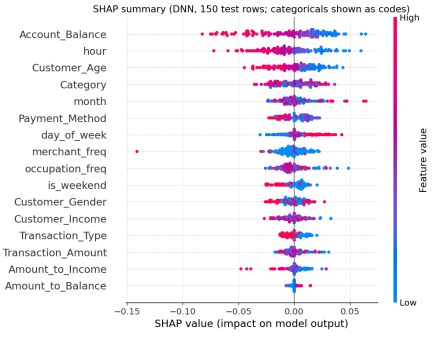
The operating DNN threshold is 0.50: the F1-maximising threshold on the validation split subject to a validation false-positive-rate guardrail of 30%. Without the guardrail, F1 is maximised at threshold 0.01, which flags 100% of transactions (F1 = 0.66 simply because the base rate is ~49%), which is the "flag everything" failure seen in an earlier version of this project. The cost-optimal validation threshold is also 0.01 (flag everything costs $6,150 vs $199,500 for approving everything, because a model with no signal cannot beat flagging everything when a missed fraud costs $500 and a false alarm $15). On the test set the cost optimum is 0.25 ($6,120). All three thresholds are recorded in models/metadata.json.



Latency: single row 6.7 ms mean / 5.8 ms p95 without SHAP; 105 ms mean / 123 ms p95 with SHAP; batch 14,159 rows/s.

5. Explainability

Top-5 SHAP factors per decision; stability 0.997 +/- 0.005 Spearman under 1% noise. Most important features: Account_Balance, hour, Customer_Age, Category, month.



6. Responsible AI

Fairness gaps (flag rate): gender 0.123, age 0.187 (criterion 0.10; not met, noisy). Robustness: DNN flips in 100% of 15 cases, median distance 1.12 SD. Human review: 267 of 809 test transactions go to MANUAL REVIEW. Full discussion in docs/RESPONSIBLE_AI_REPORT.md.

7. Deployment readiness

check	criterion	measured	status
DNN predictive power	test ROC-AUC >= 0.70	0.524	FAIL

check	criterion	measured	status
LR predictive power	test ROC-AUC >= 0.70	0.510	FAIL
Flag rate sanity	DNN flag rate <= 50%	28.2%	PASS
False-positive rate	DNN FPR <= 30%	26.8%	PASS
Expected cost vs flag-all baseline	DNN cost < flag-everything cost	$142,150 vs $6,150	FAIL
Single-row latency (no SHAP)	p95 <= 100 ms	5.8 ms	PASS
Single-row latency (with SHAP)	p95 <= 2000 ms	123 ms	PASS
Batch throughput	>= 100 rows/s	14159 rows/s	PASS
Explanation stability	mean Spearman >= 0.80	0.997 +/- 0.005	PASS
Fairness (gender)	flag-rate gap <= 0.10	0.123	FAIL
Fairness (age band)	flag-rate gap <= 0.10	0.187	FAIL
Robustness	median flip distance >= 0.5 std	1.124 std	PASS
Data drift (train vs test)	max PSI <= 0.25	0.161	PASS
Automated tests	all pass	13/13 passed	PASS
Model artifacts & metadata	all files present, metadata complete	complete	PASS

5 of 15 checks fail (predictive power, cost vs flag-all, fairness). The software checks (latency, throughput, stability, drift, tests, artifacts) pass.

8. Limitations

No learnable signal in the data; unrealistic ~49% fraud prevalence (thresholds need recalibration on realistic traffic); likely synthetic data; small fairness groups; train-vs-test drift only; SHAP explains the DNN only.

9. Future enhancements

Use real labelled data with realistic prevalence; recalibrate thresholds and probabilities; add temporal validation; richer behavioural features (velocity, device, geography); fairness mitigation; scheduled PSI monitoring job and an analyst feedback loop; authentication and audit logging.